

```
In [1]: import math
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: def plot_theta(f, g, cp, cpp, n_0, min_x, max_x):
    xs = np.arange(min_x, max_x, 1, dtype='int64')
    fn_ys = f(xs)
    gn_ysp = cp*g(xs)
    gn_yspp = cpp*g(xs)
    plt.axvline(x = n_0, color = 'k', linestyle='--')
    plt.scatter(x=xs, y=fn_ys, marker='x')
    plt.scatter(x=xs, y=gn_ysp, marker='.')
    plt.scatter(x=xs, y=gn_yspp, marker='.')
    plt.xlabel('n')
    plt.ylabel('$\lambda(n)$')
    plt.legend(['n_0 = '+str(n_0), 'f(n)', 'c\g(n)', 'c\'\'g(n)'])

def plot_oh_incr(f, g, c, n_0, min_x, max_x, incr):
    xs = np.arange(min_x, max_x, incr, dtype='int64')
    fn_ys = f(xs)
    gn_ys = c*g(xs)
    plt.axvline(x = n_0, color = 'k', linestyle='--')
    plt.scatter(x=xs, y=fn_ys, marker='x')
    plt.scatter(x=xs, y=gn_ys, marker='.')
    plt.xlabel('n')
    plt.ylabel('$\lambda(n)$')
    plt.legend(['n_0 = '+str(n_0), 'f(n)', 'cg(n)'])

def plot_oh(f, g, c, n_0, min_x, max_x):
    plot_oh_incr(f, g, c, n_0, min_x, max_x, 1)
```

Big-Oh Definition

Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $O(g(n))$ if and only if there exists positive constants c and n_0 such that:

$$f(n) \leq c \cdot g(n), \forall n \geq n_0$$

Big-Oh Using Rules

Drop smaller terms rule:

If $f(n) = (1 + h(n))$ and $h(n) \rightarrow 0$ as $n \rightarrow \infty$, then $f(n)$ is $O(1)$

Q1. Prove that $n^3 + 2n^2$ is $O(n^3)$ using the multiplication and drop smaller terms rules

Solution

- $n^3 + 2n^2$

Take n^3 outside of the brackets and simplify:

- $n^3(1 + \frac{2n^2}{n^3})$
- $n^3(1 + \frac{2}{n})$

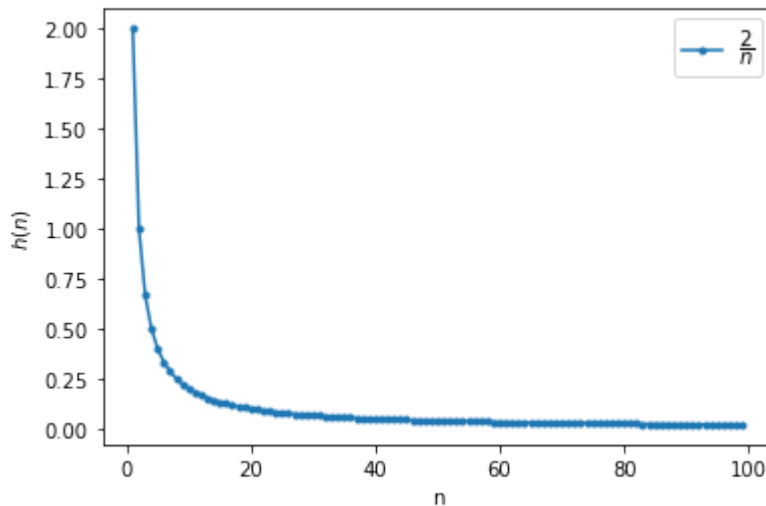
$\frac{2}{n} \rightarrow 0$ as $n \rightarrow \infty$ - see the plot below. Therefore, by the *drop smaller terms* rule, we get:

- $n^3(1) = O(n^3)$

```
In [3]: xs = np.arange(1,100,1, dtype='int64')
h = lambda n: 2 / (n)
ys = h(xs)
```

```
plt.plot(xs,ys, marker='.')
plt.xlabel('n')
plt.ylabel('$h(n)$')
plt.legend(['$\dfrac{2}{n}$'])
```

Out[3]: <matplotlib.legend.Legend at 0x11008b040>



Q2. Prove that $n^3 + 2n^2 \log(n)$ is $O(n^3)$ using the multiplication and drop smaller terms rules

Solution

- $n^3 + 2n^2 \log(n)$

Take n^3 outside of the brackets and simplify:

- $n^3(1 + \frac{2n^2 \log(n)}{n^3})$
- $n^3(1 + \frac{2 \log(n)}{n})$

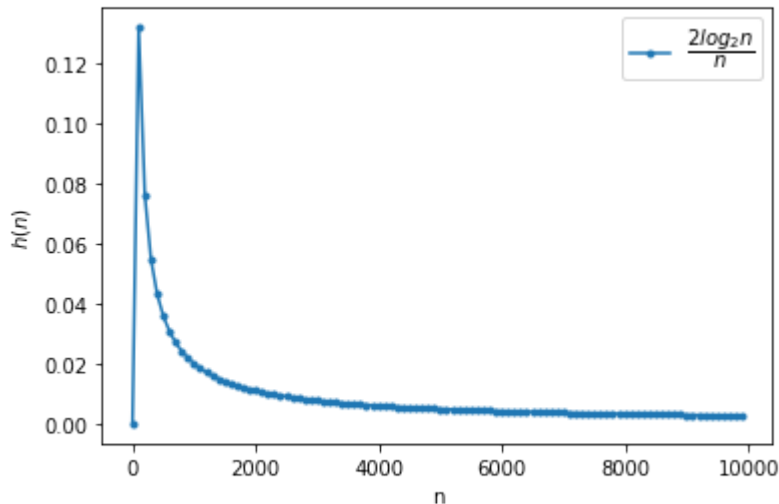
$\frac{2 \log(n)}{n} \rightarrow 0$ as $n \rightarrow \infty$ - see the plot below. Therefore, by the *drop smaller terms* rule, we get:

- $n^3(1) = O(n^3)$

```
In [4]: xs = np.arange(1,10000,100, dtype='int64')
h = lambda n: (2*np.log2(n)) / (n)
ys = h(xs)

plt.plot(xs,ys,marker='.')
plt.xlabel('n')
plt.ylabel('$h(n)$')
plt.legend(['$\frac{2\log_2 n}{n}$'])
```

Out[4]: <matplotlib.legend.Legend at 0x1101eb760>



Big-Omega and Big-Theta Definitions

Big-Omega: Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $\Omega(g(n))$ if and only if there exists strictly positive constants c and n_0 such that:

$$f(n) \geq c \cdot g(n), \forall n \geq n_0$$

Ω expresses that a function $f(n)$ grows at least as fast as $g(n)$.

Big-Theta: Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $\Theta(g(n))$ if and only if there exists positive constants c' , c'' and n_0 such that:

$$f(n) \leq c' \cdot g(n), f(n) \geq c'' \cdot g(n), \forall n \geq n_0$$

Θ expresses that a function $f(n)$ grows exactly as fast as $g(n)$.

Q3. Prove that $2n + 1$ is $\Omega(3n)$ and hence $2n + 1$ is $\Theta(3n)$

Solution

We start by proving $f(n)$ is $\Omega(3n)$:

We can say that $2n + 1$ is $\Omega(3n)$ if and only if there exists strictly positive constants c and n_0 such that $2n + 1 \geq c \cdot 3n, \forall n \geq n_0$.

We are allowed to choose any real value of $c > 0$.

The trick is to choose a fractional value of c before we simplify.

- $2n + 1 \geq c \cdot 3n, \forall n \geq n_0$
- $2n + 1 \geq \frac{1}{3} \cdot 3n, \forall n \geq n_0$ (here I am trying to make the $3n$ smaller than $2n$)
- $2n + 1 \geq n, \forall n \geq n_0$ (subtract n)
- $n + 1 \geq 0, \forall n \geq n_0$ (pick $n_0 = 1$)
- $n + 1 \geq 0, \forall n \geq 1$ (trivially true)
- $\therefore 2n + 1$ is $\Omega(3n)$ with $c = \frac{1}{3}$ and $n_0 = 1$.

Then continue with $f(n)$ is $\Theta(n)$ using c'' and n_0 from the proof for Omega:

Big-Theta: Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $\Theta(g(n))$ if and only if there exists positive constants c', c'' and n_0 such that:

$$f(n) \leq c' \cdot g(n), f(n) \geq c'' \cdot g(n), \forall n \geq n_0$$

...which gives: $2n + 1 \leq c' \cdot 3n, 2n + 1 \geq \frac{1}{3} \cdot 3n, \forall n \geq 1$

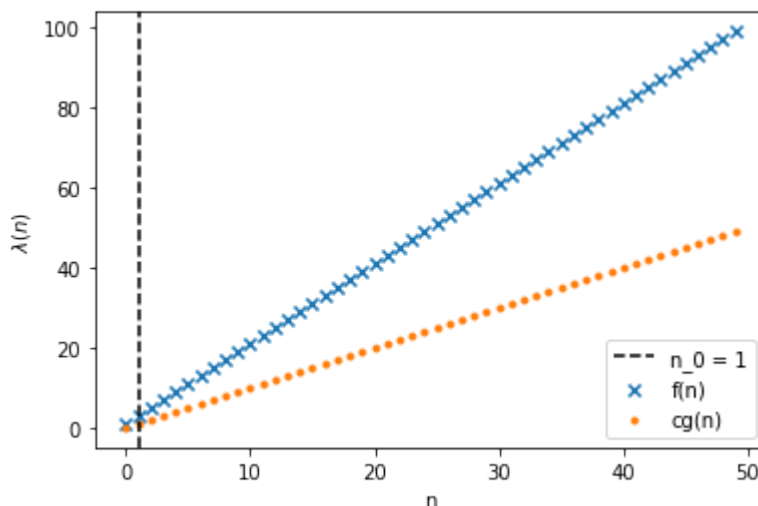
...which can be trivially satisfied with $c' = 1$.

$$2n + 1 \leq 3n, 2n + 1 \geq n, \forall n \geq 1$$

$\therefore 2n + 1$ is $\Theta(3n)$ using $c' = 1, c'' = \frac{1}{3}, n_0 = 1$.

Plot Ω

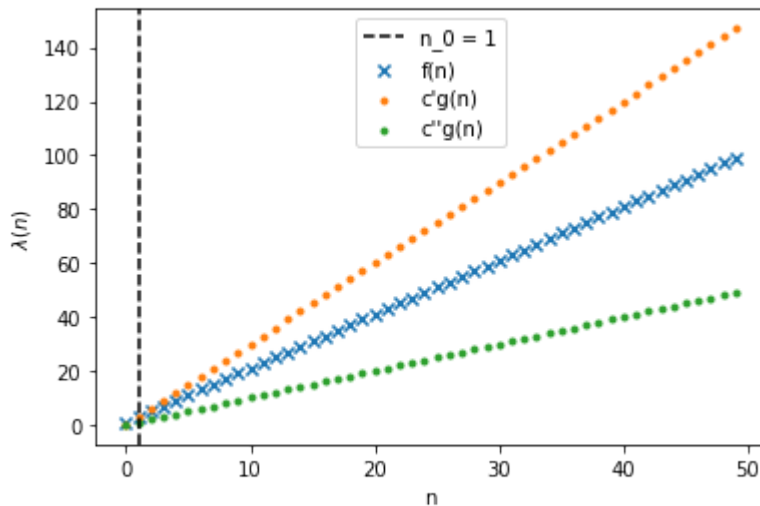
```
In [5]: c = 1/3.0
n_0 = 1
f = lambda n: (2*n+1)
g = lambda n: (3*n)
plot_oh(f, g, c, n_0, 0, 50)
```



Plot Θ

```
In [6]: cp = 1
cpp = 1/3.0
n_0 = 1
f = lambda n: 2*n+1
g = lambda n: 3*n
```

```
plot_theta(f, g, cp, cpp, n_0, 0, 50)
```



Big-Omega/Big-Theta Questions: Basics

Q4. Prove that 5 is $\Omega(1)$, and hence 5 is $\Theta(1)$

Solution

We can say that 5 is $\Omega(1)$ if and only if there exists strictly positive constants c and n_0 such that $5 \geq c \cdot 1, \forall n \geq n_0$.

- $5 \geq c \cdot 1, \forall n \geq n_0$
- $5 \geq 1, \forall n \geq n_0$ (choose a $c \leq 5$, e.g. $c = 1$)
- $5 \geq 1, \forall n \geq 1$ (which is always true, choose an arbitrary value for n_0)

$\therefore$ 5 is $\Omega(1)$ using $c = 1, n_0 = 1$.

Using $c'' = c$ and the same n_0 we just need to show the $\leq$ case for Θ .

- $5 \leq c' \cdot 1, \forall n \geq 1$
- $5 \leq 5, \forall n \geq 1$ (choose $c' = 5$)

$\therefore$ 5 is also $\Theta(1)$ using $c' = 5, c'' = 1, n_0 = 1$.

Q5. Prove that 4 is $\Omega(2)$, and hence 4 is $\Theta(2)$

Solution

We can say that 4 is $\Omega(2)$ if and only if there exists strictly positive constants c and n_0 such that $4 \geq c \cdot 2, \forall n \geq n_0$.

- $4 \geq c \cdot 2, \forall n \geq n_0$
- $4 \geq 1 \cdot 2, \forall n \geq n_0$ (choose $c = 1$)
- $4 \geq 2, \forall n \geq n_0$
- $4 \geq 2, \forall n \geq 1$ (choose an arbitrary value for n_0)

$\therefore 4$ is $\Omega(2)$

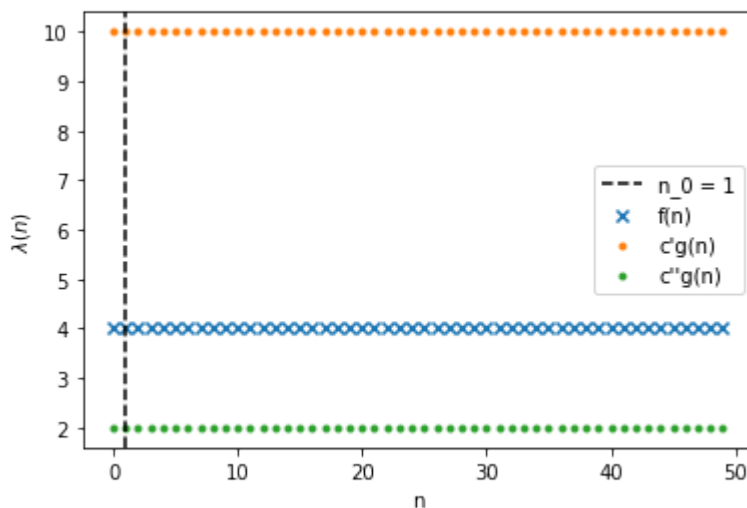
Using $c'' = c$, we just need to show the $\leq$ case for Θ .

- $4 \leq c' \cdot 2, \forall n \geq 1$
- $4 \leq 5 \cdot 2, \forall n \geq 1$ (choose a large enough value for c')
- $4 \leq 10, \forall n \geq 1$

$\therefore 4$ is also $\Theta(2)$ using $c' = 5, c'' = 1, n_0 = 1$

```
In [7]: cp = 5
cpp = 1
n_0 = 1
f = lambda n: (n*0+4)
g = lambda n: (n*0+2)

plot_theta(f, g, cp, cpp, n_0, 0, 50)
```



Q6. Prove that $2n + 1$ is $\Omega(n)$, and hence $2n + 1$ is $\Theta(n)$

Solution

We can say that $2n + 1$ is $\Omega(n)$ if and only if there exists strictly positive constants c and n_0 such that $2n + 1 \geq c \cdot n, \forall n \geq n_0$.

- $2n + 1 \geq c \cdot n, \forall n \geq n_0$

Choose $c = 1$ and pick $n_0 = 1$.

- $2n + 1 \geq n, \forall n \geq 1$
- $n + 1 \geq 0, \forall n \geq 1$

$\therefore 2n + 1$ is $\Omega(n)$

Now to complete the question!

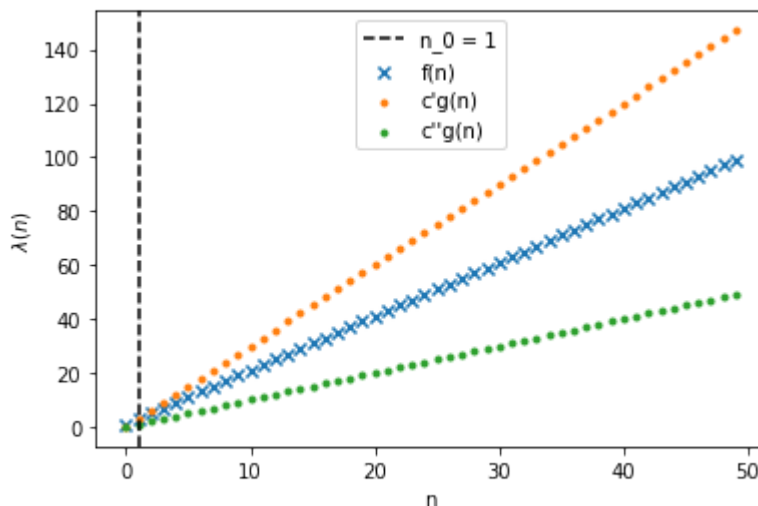
- $2n + 1 \leq c' \cdot n, \forall n \geq 1$
- $2n + 1 \leq 3n, \forall n \geq 1$ (choose $c' = 3$)

- $1 \leq n, \forall n \geq 1$

$\therefore 2n + 1$ is also $\Theta(n)$ with $c' = 3, c'' = 1, n_0 = 1$

```
In [8]: cp = 3
        cpp = 1
        n_0 = 1
        f = lambda n: (2*n+1)
        g = lambda n: (n)

        plot_theta(f, g, cp, cpp, n_0, 0, 50)
```



Big-Omega/Big-Theta Questions: Medium Difficulty

Q7. Prove that n^2 is $\Omega(2n^2)$

Solution

Definition

Given the functions $f(n)$ and $g(n)$, $f(n)$ is $\Omega(g(n))$ if there exists strictly positive constants c and n_0 such that $f(n) \geq c \cdot g(n), \forall n \geq n_0$.

Proof

- $n^2 \geq c \cdot 2n^2, \forall n \geq n_0$
- $n \geq c \cdot 2n, \forall n \geq n_0$

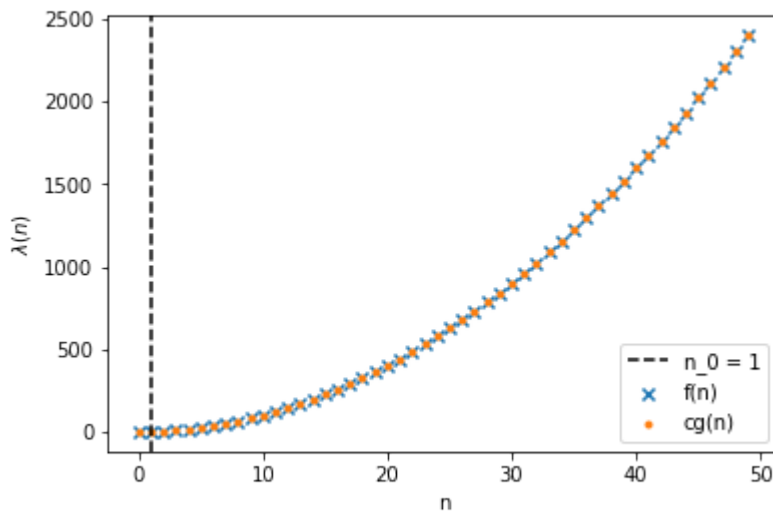
Get rid of the 2 by choosing $c = 0.5$

- $n \geq \frac{1}{2}2n, \forall n \geq n_0$
- $n \geq n, \forall n \geq n_0$
- $n \geq n, \forall n \geq 1$ (trivial, remember to pick an n_0)

```
In [9]: c = 0.5
        n_0 = 1
```

```
f = lambda n: n**2
g = lambda n: 2*n**2

plt.figure()
plot_oh(f, g, c, n_0, 0, 50)
```



Q8. Prove that $n^2 - 3$ is $\Omega(n^2)$

Solution

Definition

Given the functions $f(n)$ and $g(n)$, $f(n)$ is $\Omega(g(n))$ if there exists strictly positive constants c and n_0 such that $f(n) \geq c \cdot g(n), \forall n \geq n_0$.

Proof

- $n^2 - 3 \geq c \cdot n^2, \forall n \geq n_0$
- $n^2 - c \cdot n^2 \geq 3, \forall n \geq n_0$

Trivially choose $c < 1$ and find a suitable n_0 .

- $n^2 - \frac{1}{2} \cdot n^2 \geq 3, \forall n \geq n_0$
- $\frac{1}{2} \cdot n^2 \geq 3, \forall n \geq n_0$
- $n^2 \geq 6, \forall n \geq n_0$

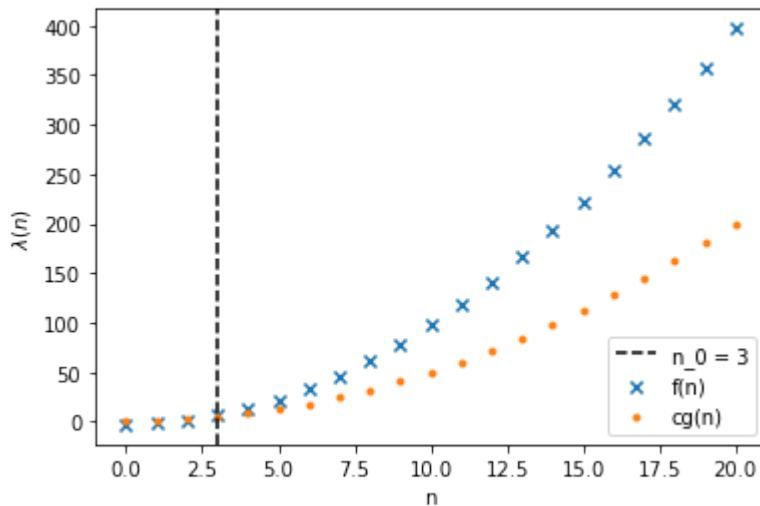
$\sqrt{6} = 2.449$, hence need to pick an $n_0 > 2.449$

- $n^2 \geq 6, \forall n \geq 3$

$\therefore n^2 - 3$ is $\Omega(n^2)$ using $c = 0.5, n_0 = 3$

```
In [10]: c = 0.5
n_0 = 3
f = lambda n: n**2-3
g = lambda n: n**2

plt.figure()
plot_oh(f, g, c, n_0, 0, 21)
```



Q9. Prove that $n^2 - 5n$ is $\Omega(n^2)$, and hence is $\Theta(n^2)$

Solution

Definition

Given the functions $f(n)$ and $g(n)$, $f(n)$ is $\Omega(g(n))$ if there exists strictly positive constants c and n_0 such that $f(n) \geq c \cdot g(n), \forall n \geq n_0$.

Proof

- $n^2 - 5n \geq c \cdot n^2, \forall n \geq n_0$
- $n - 5 \geq c \cdot n, \forall n \geq n_0$
- $n - c \cdot n \geq 5, \forall n \geq n_0$
- $\frac{1}{2}n \geq 5, \forall n \geq n_0$ (Choose $c < 1$ for example $c = 0.5$)
- $\frac{1}{2}n \geq 5, \forall n \geq 10$ (Rearranging gives $n \geq 10$, choose $n_0 = 10$)

$\therefore n^2 - 5n$ is $\Omega(n^2)$ using $c = 0.5, n_0 = 10$

Need to show for Big-Theta!

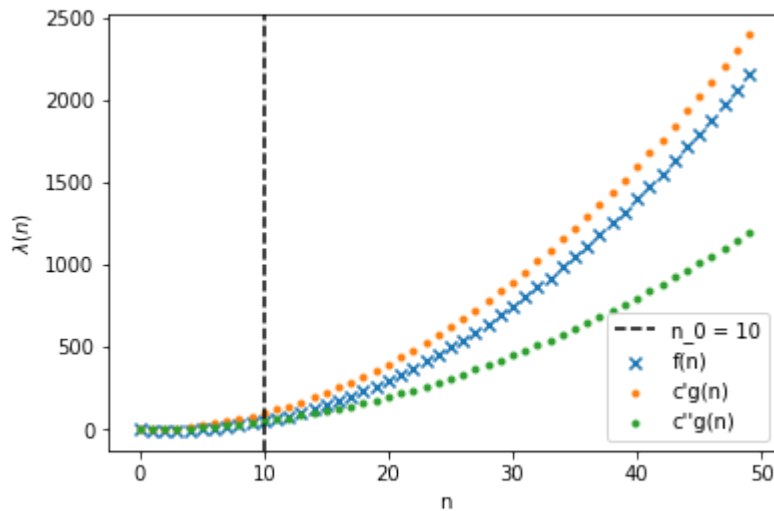
Using $c'' = 0.5$ and $n_0 = 10$ we just need to show that $n^2 - 5n \leq c' \cdot n^2, \forall n \geq 10$

- $n^2 - 5n \leq c' \cdot n^2, \forall n \geq 10$
- $n - 5 \leq c' \cdot n, \forall n \geq 10$ (divide by n)
- $n \leq n + 5, \forall n \geq 10$ (add 5 to both sides and choose $c' = 1$)
- $0 \leq 5, \forall n \geq 10$ (subtract n)

$\therefore n^2 - 5n$ is also $\Theta(n^2)$ using $c' = 1, c'' = 0.5, n_0 = 10$

```
In [11]: cp = 1
cpp = 0.5
n_0 = 10
f = lambda n: n**2-5*n
g = lambda n: n**2

plot_theta(f, g, cp, cpp, n_0, 0, 50)
```



Q10. Prove that $n^2 + 1$ is $\Omega(n^2)$

Solution

Definition

Given the functions $f(n)$ and $g(n)$, $f(n)$ is $\Omega(g(n))$ if there exists strictly positive constants c and n_0 such that $f(n) \geq c \cdot g(n), \forall n \geq n_0$.

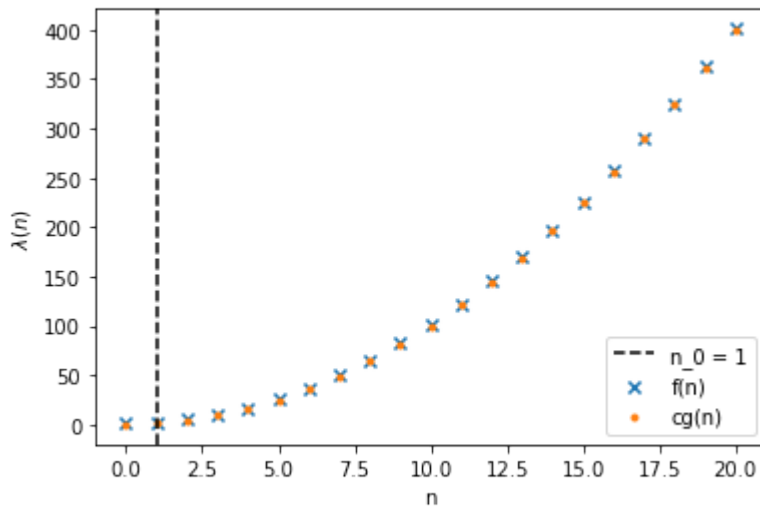
Proof

- $n^2 + 1 \geq c \cdot n^2, \forall n \geq n_0$
- $n^2 + 1 \geq n^2, \forall n \geq n_0$
- $1 \geq 0, \forall n \geq 1$

$\therefore n^2 + 1$ is $\Omega(n^2)$ using $c = 1, n_0 = 1$

```
In [12]: c = 1
n_0 = 1
f = lambda n: n**2+1
g = lambda n: n**2

plt.figure()
plot_oh(f, g, c, n_0, 0, 21)
```



Little-Oh Definition

Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $o(g(n))$ if **for all positive real constants** $c > 0$ there exists n_0 such that:

$$f(n) < c \cdot g(n), \forall n \geq n_0$$

Important: $c \in \mathbb{R}_*^+$ where $\mathbb{R}_*^+ = \{x \in \mathbb{R} | x > 0\}$

Q11. Prove or disprove that 5 is $o(1)$

Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $o(g(n))$ if **for all positive real constants** $c > 0$ there exists n_0 such that $f(n) < c \cdot g(n), \forall n \geq n_0$.

- $5 < c \cdot 1, \forall c > 0, \exists n_0 > 0, \forall n > n_0$

This is not valid for all $0 < c < 5$ hence it is disproven that 5 is $o(1)$.

Q12. Prove or disprove that 5 is $o(n)$

Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $o(g(n))$ if **for all positive real constants** $c > 0$ there exists n_0 such that $f(n) < c \cdot g(n), \forall n \geq n_0$.

- $5 < c \cdot n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $\frac{5}{c} < n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$

Since we have $\forall c > 0, \exists n_0 > 0$ we can pick an n_0 that depends on c . Since we have that $n > \frac{5}{c}$, trivially choose $n_0 = \frac{5}{c} + 1$

- $\frac{5}{c} < n, \forall c > 0, \exists n_0 > 0, \forall n > \frac{5}{c} + 1$

$\therefore$ 5 is $o(n)$ using $n_0 = \frac{5}{c} + 1$

Little-Oh Questions

Q13. Prove or disprove that n is $o(n^2)$

Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $o(g(n))$ if **for all positive real constants** $c > 0$ there exists n_0 such that $f(n) < c \cdot g(n), \forall n \geq n_0$.

- $n < c \cdot n^2, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $1 < c \cdot n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$ (divide both sides by n)
- $\frac{1}{c} < n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$ (divide both sides by c)

Since we have $\forall c > 0, \exists n_0 > 0$ we can pick an n_0 that depends on c . Since we have that $n > \frac{1}{c}$, trivially choose $n_0 = \frac{1}{c} + 1$

- $\frac{1}{c} < n, \forall c > 0, \exists n_0 > 0, \forall n > \frac{1}{c} + 1$

$\therefore n$ is $o(n^2)$ using $n_0 = \frac{1}{c} + 1$

Q14. Prove or disprove that 1 is $o(\log n)$

Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $o(g(n))$ if **for all positive real constants** $c > 0$ there exists n_0 such that $f(n) < c \cdot g(n), \forall n \geq n_0$.

- $1 < c \cdot \log n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $\frac{1}{c} < \log n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$

To isolate n on the right hand side, we can do 2^x for both sides - this allows us to eliminate the $\log$ on the right. Remember that if the log base is not specified, assume we are using a base of 2.

- $2^{\frac{1}{c}} < 2^{\log n}, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $2^{\frac{1}{c}} < n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$

Since we have $\forall c > 0, \exists n_0 > 0$ we can pick an n_0 that depends on c . Since we have that $n > 2^{\frac{1}{c}}$, trivially choose $n_0 = 2^{\frac{1}{c}} + 1$

- $2^{\frac{1}{c}} < n, \forall c > 0, \exists n_0 > 0, \forall n > 2^{\frac{1}{c}} + 1$

$\therefore 1$ is $o(\log n)$ using $n_0 = 2^{\frac{1}{c}} + 1$

Q15. Prove or disprove that $\log n$ is $o(1)$

Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $o(g(n))$ if **for all positive real constants** $c > 0$ there exists n_0 such that $f(n) < c \cdot g(n), \forall n \geq n_0$.

- $\log n < c \cdot 1, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $\log n < c, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $2^{\log n} < 2^c, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $n < 2^c, \forall c > 0, \exists n_0 > 0, \forall n > n_0$

No matter the value we choose for n_0 , for any $n > 2^c$ the inequality will fail. We therefore don't satisfy the condition that it must work $\forall n > n_0$, hence it is disproven that $\log n$ is $o(1)$.

Additional Practice Questions (more challenging)

Q16. Prove or disprove that 1 is $\Omega(n)$

- $1 \geq c \cdot n, \forall n \geq n_0$

Choose $c = 1$

- $1 \geq n, \forall n \geq n_0$

Here, 1 ends up as an upperbound on the value of n . Any value chosen for c and n_0 will still lead to this inequality failing for larger values of n . We therefore don't satisfy the condition that it must work $\forall n > n_0$, hence it is disproven.

Q17. Prove or disprove that n is $\Omega(1)$

- $n \geq c \cdot 1, \forall n \geq n_0$
- $n \geq 1, \forall n \geq n_0$
- $n \geq 1, \forall n \geq 1$

$\therefore n$ is $\Omega(1)$ using $c = 1, n_0 = 1$

Q18. Prove or disprove that n^2 is $\Omega(n)$

- $n^2 \geq c \cdot n, \forall n \geq n_0$
- $n \geq c, \forall n \geq n_0$
- $n \geq 1, \forall n \geq n_0$
- $n \geq 1, \forall n \geq 1$

$\therefore n^2$ is $\Omega(n)$ using $c = 1, n_0 = 1$

Q19. Prove or disprove that n is $o(n \log n)$

Given positive functions $f(n)$ and $g(n)$, we can say that $f(n)$ is $o(g(n))$ if **for all positive real constants** $c > 0$ there exists n_0 such that $f(n) < c \cdot g(n), \forall n \geq n_0$.

- $n < c \cdot n \log n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $1 < c \cdot \log n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $\frac{1}{c} < \log n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$

To isolate n on the right hand side, we can do 2^x for both sides - this allows us to eliminate the $\log$ on the right. Remember that if the $\log$ base is not specified, assume we are using a base of 2.

- $2^{\frac{1}{c}} < 2^{\log n}, \forall c > 0, \exists n_0 > 0, \forall n > n_0$
- $2^{\frac{1}{c}} < n, \forall c > 0, \exists n_0 > 0, \forall n > n_0$

Since we have $\forall c > 0, \exists n_0 > 0$ we can pick an n_0 that depends on c . Since we have that $n > 2^{\frac{1}{c}}$, trivially choose $n_0 = 2^{\frac{1}{c}} + 1$

- $2^{\frac{1}{c}} < n, \forall c > 0, \exists n_0 > 0, \forall n > 2^{\frac{1}{c}} + 1$

$\therefore n$ is $o(n \log n)$ using $n_0 = 2^{\frac{1}{c}} + 1$

Q20. Given that $f(n) = n^2$ if n is even, and $f(n) = n$ if n is odd. From the definitions, find the O and Ω behaviours of $f(n)$.

Warning: be careful to find a single c that works for all n , not separate c for even and odd n

Big-Oh is the worst case, and for $f(n)$ the worst case is " n^2 if n is even". So we prove $f(n)$ is $O(n^2)$.

- $f(n) \leq c \cdot n^2, \forall n \geq n_0$
- Even case: $n^2 \leq c \cdot n^2, \forall n \geq n_0$
- Odd case: $n \leq c \cdot n^2, \forall n \geq n_0$

Choose $c = 1$:

- Even case: $n^2 \leq n^2, \forall n \geq n_0$
- Odd case: $n \leq n^2, \forall n \geq n_0$

Both of these statements are trivially true, so we can just choose $n_0 = 1$.

$\therefore f(n)$ is $O(n^2)$.

You may then assume it is also $\Omega(n^2)$; however, you would need to find some c and n_0 such that $f(n) \geq c \cdot n^2, \forall n \geq n_0$.

But this would fail for the cases where n is odd.

It is important to understand the difference between this and the function $n^2 + n$, which is $\Omega(n^2)$.

Instead, we can prove $f(n)$ is $\Omega(n)$:

- $f(n) \geq c \cdot n, \forall n \geq n_0$
- Even case: $n^2 \geq c \cdot n, \forall n \geq n_0$
- Odd case: $n \geq c \cdot n, \forall n \geq n_0$

Choose $c = 1$:

- Even case: $n^2 \geq n, \forall n \geq n_0$
- Odd case: $n \geq n, \forall n \geq n_0$

Again, both of these statements are trivially true, so we can just choose $n_0 = 1$.

$\therefore f(n)$ is $\Omega(n)$.

Q21. Prove or disprove that $n \log n$ is $o(n^2)$